

Test: Unit 2
Reactions

Name: _____

Instructions:

You will have the entire period to complete the test (75 minutes)

The test is made up of 4 sections:

Section A – Multiple Choice

Section B – Matching

Section C – short answers

Section D – Equations

Please follow all the instructions

Please read all questions carefully

A mark breakdown of the test can be found after these instructions

Good Luck!

Mark Summary

Section	Knowledge /Understanding	Thinking	Application	Communication	TOTAL
A	10	/	/	/	10
B	10	/	/	/	10
C	/	/	4	2	6
D	/	19	/	6	25
Sub-Total	/ 20	/ 19	/ 4	/ 8	/ 51

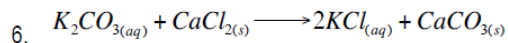
Comments:

Section A – Multiple Choice (10 Marks – Knowledge/Understanding)

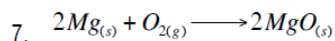
Choose the **best** answer for each of the following.

1. Which of the following would be the best example of a chemical change?
 - a. Burning toast
 - b. Melting water
 - c. Squeezing a sponge
 - d. Adding more sugar to a glass of water
 - e. Shaking salad dressing so that the oil and vinegar mix
2. Which of the following components are not necessary for every for every balanced chemical equation?
 - a. Equal amounts of reactants and products
 - b. Proper chemical formulas
 - c. Coefficients in front of every compound
 - d. The states of every compound
 - e. Multiple products
3. When carbon dioxide is allowed to react with water, the resulting solution would be:
 - a. A salt
 - b. Neutral
 - c. Basic
 - d. Acidic
 - e. A very poor conductor of electricity
4. When magnesium oxide is allowed to react with water, the resulting solution would be:
 - a. A salt
 - b. Neutral
 - c. Basic
 - d. Acidic
 - e. A very poor conductor of electricity
5. Indicate which of the following compounds would not be soluble in water.
 - a. Sodium Acetate
 - b. Ferric Bromide
 - c. Ammonium Hydroxide
 - d. Potassium Sulfate
 - e. Lead II Carbonate

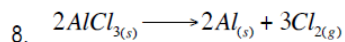
For the next five (5) questions, choose the correct name for the given reactions.



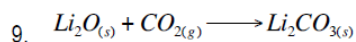
- a. Synthesis
- b. Decomposition
- c. Combustion
- d. Single Displacement
- e. Double Displacement



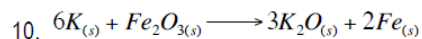
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Section B – Matching (10 Marks – K/U)

Match the term on the left with the appropriate definition on the right.

	1	Product	A	This occurs when an equal amount of acid solution is mixed with an equal amount of basic solution
	2	Reactant	B	This type of reaction could involve the formation of one larger compound from two smaller ones.
	3	Water	C	These types of reactions will often produce a solid precipitate.
	4	Neutralization	D	A compound that has formed after a chemical reaction has taken place.
	5	Antoine Lavoisier	E	A Hoffman apparatus could be used to perform this type of reaction with water as a reactant.
	6	Hydrocarbon	F	This is an organic compound that only contains carbon and oxygen.
	7	Synthesis Reaction	G	A compound or element that exists before a chemical reaction occurs.
	8	Single Displacement Reaction	H	He created the first metal activity series.
	9	Double Displacement Reaction	I	We often use the metal activity series to determine if this type of reaction will proceed.
	10	Decomposition Reaction	J	This common oxide forms after hydrogen has been involved in a combustion reaction.
			K	He was able to develop the law of conservation of mass through very precise experimentation.
			L	This is an organic compound that only contains hydrogen and carbon

Section C – Short Answer (6 Marks)

- Carbon dioxide is a very common oxide that is formed as a result of a combustion reaction. Briefly explain how this oxide is able to affect the pH of rainwater. (2 Marks [A])
- Answer the following questions about combustion:
 - Using a simple word equation or skeleton equation, explain the difference between complete and incomplete combustion (2 Marks [C])
 - One of the products of the incomplete combustion of a hydrocarbon is CO. Why is CO harmful to humans? (2 Marks [A])

Section D – Working with Chemical Equations (25 Marks)

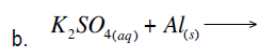
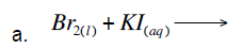
1. Create balanced chemical equations for each of the following:
 - a. Doctors can sometimes treat acid indigestion by prescribing solid magnesium hydroxide (milk of magnesia). The milk of magnesia reacts with the excess stomach acid (hydrochloric acid – aqueous) to form water and an aqueous ionic compound. (3 Marks [C])
 - b. Acids cannot be poured down the drain because they can react with certain salts to form potentially deadly gases. For example, hydrogen sulfide gas (a poisonous, rotten-egg smelling gas) is formed along with aqueous sodium chloride when aqueous hydrochloric acid is poured down the drain and makes contact with solid sodium sulfide. (3 Marks [C])

2. Predict the products formed by reactions in aqueous solutions of each of the following pairs of compounds. In each case, write a balanced chemical equation indicating the physical state of the products formed. (6 Marks [T/I])

a. Copper II Nitrate and Magnesium Chloride.

b. Ammonium Sulfide and Iron II Sulfate.

3. For each of the following, use the activity series to determine which reactions will proceed. For the reactions that do occur, predict the products and complete and balance the equation. For the reactions that will not occur, write "no reaction." (6 Marks [T/I])



4. Determine which of the following reactions will produce a precipitate. For those that do, write both a **balanced chemical equation** and a **net ionic equation**. If no precipitate is produced, write "no precipitate." (7 Marks [T/I])

